ให้

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix},\quad B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix},\quad a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix},\quad b = \begin{bmatrix} 3 & 0 & 8 \end{bmatrix}.$$

จงคำนวณผลคูณและผลบวกต่อไปนี้ หรือให้เหตุผลว่าทำไมถึงหาค่าไม่ได้ (Undefined)

1. $A\mathrm{b}^T + B\mathrm{b}^T, (A+B)b^T, bA, BA, B - B^T$
2. $A^T b, b^T B, (3A - 2B)^T a, a^T(3A - 2B)$
3. $ab - ba, -(4b)(7a), -28ba, 5abB$
4. $B^3, BC, (BC)^2, (BC)(BC)^T$
5. $IA, AI, A(I + A), A + IA$

คำตอบ

1. $\begin{bmatrix} \quad & \quad & \quad \end{bmatrix}, \begin{bmatrix} \quad & \quad & \quad \end{bmatrix}, \ldots$
2.
3.
4.

Solutions and detailed **explanations** are provided on the **subsequent** pages.

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BY

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Problem 1

Given $A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$, $b = \begin{bmatrix} 3 & 0 & 8 \end{bmatrix}$

Find 1) $A\mathrm{b}^T + B\mathrm{b}^T$

2) $(A + B)b^T$

3) bA

4) BA

5) $B - B^T$

Problem 1.1

$$A\mathrm{b}^T + B\mathrm{b}^T$$

First, find b^T

$$b = \begin{bmatrix} 3 & 0 & 8 \end{bmatrix}$$

$$\mathrm{b}^T = \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix}$$

Next, Compute $A \times b^T$

$$Ab^T = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} (6 \times 3) + (-2 \times 0) + (2 \times 8) \\ (10 \times 3) + (-3 \times 0) + (1 \times 8) \\ (-10 \times 3) + (5 \times 0) + (1 \times 8) \end{bmatrix} = \begin{bmatrix} 34 \\ 38 \\ -22 \end{bmatrix}$$

$$= \begin{bmatrix} 34 \\ 38 \\ -22 \end{bmatrix}$$

Then, Compute $B \times b^T$

$$Bb^T = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} (9 \times 3) + (4 \times 0) + (-4 \times 8) \\ (4 \times 3) + (-7 \times 0) + (0 \times 8) \\ (-4 \times 3) + (0 \times 0) + (11 \times 8) \end{bmatrix} = \begin{bmatrix} -5 \\ 12 \\ 76 \end{bmatrix}$$

$$= \begin{bmatrix} -5 \\ 12 \\ 76 \end{bmatrix}$$

Thus,

$$A\mathrm{b}^T + B\mathrm{b}^T = \begin{bmatrix} 34 \\ 38 \\ -22 \end{bmatrix} + \begin{bmatrix} -5 \\ 12 \\ 76 \end{bmatrix}$$

$$= \begin{bmatrix} 34 + (-5) \\ 38 + 12 \\ -22 + 76 \end{bmatrix} = \begin{bmatrix} 29 \\ 50 \\ 54 \end{bmatrix}$$

$$= \begin{bmatrix} 29 \\ 50 \\ 54 \end{bmatrix}$$

Commented [PP1]: Answer 1.1

Problem 1.2

$$(A+B)b^T$$

First, find $(A+B)$

$$A+B=\begin{bmatrix}6 & -2 & 2\\ 10 & -3 & 1\\ -10 & 5 & 1\end{bmatrix}+\begin{bmatrix}9 & 4 & -4\\ 4 & 7 & 0\\ -4 & 0 & 11\end{bmatrix}$$
$$=\begin{bmatrix}6+9 & (-2)+4 & 2+(-4)\\ 10+4 & (-3)+7 & 1+0\\ -10+4 & 5+0 & 1+11\end{bmatrix}=\begin{bmatrix}15 & 2 & -2\\ 14 & 4 & 1\\ -6 & 5 & 12\end{bmatrix}$$
$$=\begin{bmatrix}15 & 2 & -2\\ 14 & 4 & 1\\ -6 & 5 & 12\end{bmatrix}$$

Next, Evaluate $(A+B)b^T$

$$(A+B)b^T=\begin{bmatrix}15 & 2 & -2\\ 14 & 4 & 1\\ -6 & 5 & 12\end{bmatrix}\begin{bmatrix}3\\ 0\\ 8\end{bmatrix}$$
$$=\begin{bmatrix}(15\times 3)+(2\times 0)+(-2\times 8)\\ (14\times 3)+(4\times 0)+(1\times 8)\\ (-6\times 3)+(5\times 0)+(12\times 8)\end{bmatrix}$$
$$=\begin{bmatrix}29\\ 50\\ 78\end{bmatrix}$$

Commented [PP2]: Answer 1.2

Problem 1.3

$$bA$$

Evaluate expression bA

$$
\begin{aligned}
bA &= \begin{bmatrix} 3 & 0 & 8 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \\
&= \begin{bmatrix} (18+0-80) & (-6+0+40) & (6+0+8) \end{bmatrix} \\
&= \begin{bmatrix} -62 & 34 & 14 \end{bmatrix}
\end{aligned}
$$

Commented [PP3]: Answer 1.3

Problem 1.4

$$BA$$

Evaluate expression BA

$$
\begin{aligned}
BA &= \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 54+40+40 & -18-12-20 & 18+4-4 \\ 24+70+0 & -8-21+0 & 8+7+0 \\ -24+0-110 & 8+0+55 & -8+0+11 \end{bmatrix} \\
&= \begin{bmatrix} 134 & -50 & 18 \\ 94 & -29 & 15 \\ -134 & 63 & 3 \end{bmatrix}
\end{aligned}
$$

Commented [PP4]: Answer 1.4

Problem 1.5

$$B - B^T$$

Calculate $B - B^T$

$$B - B^T = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} - \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$
$$= \begin{bmatrix} 9-9 & 4-4 & -4-(-4) \\ 4-4 & 7-7 & 0-0 \\ -4-(-4) & 0-0 & 11-11 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Commented [PP5]: Answer 1.5

Problem 2

Given

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}, \quad b = \begin{bmatrix} 3 & 0 & 8 \end{bmatrix}$$

Find

1) $A^T b$
2) $b^T B$
3) $(3A - 2B)^T a$
4) $a^T \ (3A - 2B)$

Problem 2.1

$$A^T b$$

First, find A^T

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 6 & 10 & -10 \\ -2 & -3 & 5 \\ 2 & 1 & 1 \end{bmatrix}$$

Next, Compute $A^T b$

A^T is a 3×3 matrix, and b is a 1×3 matrix.

Since the inner dimensions (3 and 1) do not match, the multiplication is $Undefined$.

Commented [PP6]: Answer 2.1

Problem 2.2

$$b^T B$$

First, find b^T

$$b = [3 \quad 0 \quad 8]$$

$$b^T = \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix}$$

Next, Compute $b^T B$

b^T is a 3×1 matrix, and B is a 3×3 matrix.

Since the inner dimensions (1 and 3) do not match, the multiplication is $Undefined$.

Commented [PP7]: 2.2

Problem 2.3

$$(3A-2B)^T a$$

First, find $3A-2B$

$$3A = 3\begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 18 & -6 & 6 \\ 30 & -9 & 3 \\ -30 & 15 & 3 \end{bmatrix}$$

$$2B = 2\begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} = \begin{bmatrix} 18 & 8 & -8 \\ 8 & 14 & 0 \\ -8 & 0 & 22 \end{bmatrix}$$

$$3A-2B = \begin{bmatrix} 18-18 & -6-8 & 6-(-8) \\ 30-8 & -9-14 & 3-0 \\ -30-(-8) & 15-0 & 3-22 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -14 & 14 \\ 22 & -23 & 3 \\ -22 & 15 & -19 \end{bmatrix}$$

Next, find $(3A-2B)^T$

$$(3A-2B)^T = \begin{bmatrix} 0 & 22 & -22 \\ -14 & -23 & 15 \\ 14 & 3 & -19 \end{bmatrix}$$

Then, Compute $(3A-2B)^T a$

$$(3A-2B)^T a = \begin{bmatrix} 0 & 22 & -22 \\ -14 & -23 & 15 \\ 14 & 3 & -19 \end{bmatrix}\begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} (0 \times 5) + (22 \times 1) + (-22 \times 2) \\ (-14 \times 5) + (-23 \times 1) + (15 \times 2) \\ (14 \times 5) + (3 \times 1) + (-19 \times 2) \end{bmatrix}$$

$$= \begin{bmatrix} 0 + 22 - 44 \\ -70 - 23 + 30 \\ 70 + 3 - 38 \end{bmatrix}$$

$$= \begin{bmatrix} -22 \\ -63 \\ 35 \end{bmatrix}$$

Commented [PP8]: Answer 2.3

Problem 2.4

$$a^T(3A - 2B)$$

First, find a^T

$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

$$a^T = \begin{bmatrix} 5 & 1 & 2 \end{bmatrix}$$

We already have $3A - 2B = \begin{bmatrix} 0 & -14 & 14 \\ 22 & -23 & 3 \\ -22 & 15 & -19 \end{bmatrix}$

Next, Compute $a^T(3A - 2B)$

$$a^T(3A - 2B) = \begin{bmatrix} 5 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 & -14 & 14 \\ 22 & -23 & 3 \\ -22 & 15 & -19 \end{bmatrix}$$

$$= \begin{bmatrix} (0 + 22 - 44) & (-70 - 23 + 30) & (70 + 3 - 38) \end{bmatrix}$$

$$= \begin{bmatrix} -22 & -63 & 35 \end{bmatrix}$$

Commented [PP9]: Answer 2.4

Problem 3

Given

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}, \ b = \begin{bmatrix} 3 & 0 & 8 \end{bmatrix}$$

Find

1) $ab - ba$
2) $-(4b)(7a)$
3) $-28ba$
4) $5abB$

Problem 3.1

$$ab - ba$$

First, compute ab

$$\begin{aligned} ab &= \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 0 & 8 \end{bmatrix} \\ &= \begin{bmatrix} (5 \times 3) & (5 \times 0) & (5 \times 8) \\ (1 \times 3) & (1 \times 0) & (1 \times 8) \\ (2 \times 3) & (2 \times 0) & (2 \times 8) \end{bmatrix} \\ &= \begin{bmatrix} 15 & 0 & 40 \\ 3 & 0 & 8 \\ 6 & 0 & 16 \end{bmatrix} \end{aligned}$$

Next, compute ba

$$
\begin{aligned}
ba &= \begin{bmatrix} 3 & 0 & 8 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \\
&= [(3 \times 5) + (0 \times 1) + (8 \times 2)] \\
&= [15 + 0 + 16] \\
&= [31]
\end{aligned}
$$

Then, calculate $ab - ba$

Since ab is a 3×3 matrix and ba is a 1×1 matrix, they have different dimensions. Therefore, subtraction is $Undefined$.

Commented [PP10]: Answer 3.1

Problem 3.2

$$-(4b)(7a)$$

First, find $4b$ and $7a$

$$
\begin{aligned}
4b &= 4\begin{bmatrix} 3 & 0 & 8 \end{bmatrix} \\
&= \begin{bmatrix} 12 & 0 & 32 \end{bmatrix} \\
7a &= 7\begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \\
&= \begin{bmatrix} 35 \\ 7 \\ 14 \end{bmatrix}
\end{aligned}
$$

Then, compute $-(4b)(7a)$

$$
\begin{aligned}
-(4b)(7a) &= -\begin{bmatrix} 12 & 0 & 32 \end{bmatrix} \begin{bmatrix} 35 \\ 7 \\ 14 \end{bmatrix} \\
&= -[(12 \times 35) + (0 \times 7) + (32 \times 14)] \\
&= -[420 + 0 + 448] \\
&= [-868]
\end{aligned}
$$

Therefore,

$-(4b)(7a) = [-868]$

Commented [PP11]: Answer 3.2

Problem 3.3

$$-28ba$$

We already have $ba = [31]$

Compute $-28ba$

$$-28ba = -28[31]$$

$= [-868]$

Commented [PP12]: Answer 3.3

Problem 3.4

$$5abB$$

We already have $ab = \begin{bmatrix} 15 & 0 & 40 \\ 3 & 0 & 8 \\ 6 & 0 & 16 \end{bmatrix}$

First, find $5ab$

$$5ab = 5\begin{bmatrix} 15 & 0 & 40 \\ 3 & 0 & 8 \\ 6 & 0 & 16 \end{bmatrix}$$

$$a = \begin{bmatrix} 75 & 0 & 200 \\ 15 & 0 & 40 \\ 30 & 0 & 80 \end{bmatrix}$$

Next, compute $(5ab)B$

$$5abB = \begin{bmatrix} 75 & 0 & 200 \\ 15 & 0 & 40 \\ 30 & 0 & 80 \end{bmatrix}\begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} (75 \times 9) + (0 \times 4) + (200 \times -4) & (75 \times 4) + (0 \times 7) + (200 \times 0) & (75 \times -4) + (0 \times 0) + (200 \times 11) \\ (15 \times 9) + (0 \times 4) + (40 \times -4) & (15 \times 4) + (0 \times 7) + (40 \times 0) & (15 \times -4) + (0 \times 0) + (40 \times 11) \\ (30 \times 9) + (0 \times 4) + (80 \times -4) & (30 \times 4) + (0 \times 7) + (80 \times 0) & (30 \times -4) + (0 \times 0) + (80 \times 11) \end{bmatrix}$$

$$= \begin{bmatrix} 675 + 0 - 800 & 300 + 0 + 0 & -300 + 0 + 2200 \\ 135 + 0 - 160 & 60 + 0 + 0 & -60 + 0 + 440 \\ 270 + 0 - 320 & 120 + 0 + 0 & -120 + 0 + 880 \end{bmatrix}$$

$$= \begin{bmatrix} -125 & 300 & 1900 \\ -25 & 60 & 380 \\ -50 & 120 & 760 \end{bmatrix}$$

Commented [PP13]: Answer 3.4

Problem 4

Given

$$B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}, C = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}$$

Find

1) B^3
2) BC
3) $(BC)^2$
4) $(BC)(BC)^T$

Problem 4.1

$$B^3$$

First, compute B^2

$$\begin{aligned} B^2 &= B \times B \\ &= \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \\ &= \begin{bmatrix} (9\times 9)+(4\times 4)+(-4\times -4) & (9\times 4)+(4\times 7)+(-4\times 0) & (9\times -4)+(4\times 0)+(-4\times 11) \\ (4\times 9)+(7\times 4)+(0\times -4) & (4\times 4)+(7\times 7)+(0\times 0) & (4\times -4)+(7\times 0)+(0\times 11) \\ (-4\times 9)+(0\times 4)+(11\times -4) & (-4\times 4)+(0\times 7)+(11\times 0) & (-4\times -4)+(0\times 0)+(11\times 11) \end{bmatrix} \\ &= \begin{bmatrix} 81+16+16 & 36+28+0 & -36+0-44 \\ 36+28+0 & 16+49+0 & -16+0+0 \\ -36+0-44 & -16+0+0 & 16+0+121 \end{bmatrix} \\ &= \begin{bmatrix} 113 & 64 & -80 \\ 64 & 65 & -16 \\ -80 & -16 & 137 \end{bmatrix} \end{aligned}$$

Next, compute $B^3 = B^2B$

$$B^3 = \begin{bmatrix} 113 & 64 & -80 \\ 64 & 65 & -16 \\ -80 & -16 & 137 \end{bmatrix} \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} (1017 + 256 + 320) & (452 + 448 + 0) & (-452 + 0 - 880) \\ (576 + 260 + 64) & (256 + 455 + 0) & (-256 + 0 - 176) \\ (-720 - 64 - 548) & (-320 - 112 + 0) & (320 + 0 + 1507) \end{bmatrix}$$

$$= \begin{bmatrix} 1593 & 900 & -1332 \\ 900 & 711 & -432 \\ -1332 & -432 & 1827 \end{bmatrix}$$

Commented [PP14]: Answer 4.1

Problem 4.2

$$BC$$

Compute $B \times C$

$$BC = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (9 \times 3) + (4 \times 0) + (-4 \times 4) & (9 \times 1) + (4 \times -2) + (-4 \times 0) \\ (4 \times 3) + (7 \times 0) + (0 \times 4) & (4 \times 1) + (7 \times -2) + (0 \times 0) \\ (-4 \times 3) + (0 \times 0) + (11 \times 4) & (-4 \times 1) + (0 \times -2) + (11 \times 0) \end{bmatrix}$$

$$= \begin{bmatrix} 27 + 0 - 16 & 9 - 8 + 0 \\ 12 + 0 + 0 & 4 - 14 + 0 \\ -12 + 0 + 44 & -4 + 0 + 0 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 1 \\ 12 & -10 \\ 32 & -4 \end{bmatrix}$$

Commented [PP15]: Answer 4.2

Problem 4.3

$$(BC)^2$$

$(BC)^2 = (BC)(BC)BC$ is a 3×2 matrix. Since we are multiplying a 3×2 matrix by another 3×2 matrix, the inner dimensions (2 and 3) do not match. Therefore, the multiplication is $Undefined$.

Commented [PP16]: Answer 4.3

Problem 4.4

$$(BC)(BC)^T$$

We already have $BC = \begin{bmatrix} 11 & 1 \\ 12 & -10 \\ 32 & -4 \end{bmatrix}$

First, find $(BC)^T$

$$(BC)^T = \begin{bmatrix} 11 & 12 & 32 \\ 1 & -10 & -4 \end{bmatrix}$$

Next, Compute $(BC)(BC)^T$

$$(BC)(BC)^T = \begin{bmatrix} 11 & 1 \\ 12 & -10 \\ 32 & -4 \end{bmatrix} \begin{bmatrix} 11 & 12 & 32 \\ 1 & -10 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} (11 \times 11) + (1 \times 1) & (11 \times 12) + (1 \times -10) & (11 \times 32) + (1 \times -4) \\ (12 \times 11) + (-10 \times 1) & (12 \times 12) + (-10 \times -10) & (12 \times 32) + (-10 \times -4) \\ (32 \times 11) + (-4 \times 1) & (32 \times 12) + (-4 \times -10) & (32 \times 32) + (-4 \times -4) \end{bmatrix}$$

$$= \begin{bmatrix} 121 + 1 & 132 - 10 & 352 - 4 \\ 132 - 10 & 144 + 100 & 384 + 40 \\ 352 - 4 & 384 + 40 & 1024 + 16 \end{bmatrix}$$

$$= \begin{bmatrix} 122 & 122 & 348 \\ 122 & 244 & 424 \\ 348 & 424 & 1040 \end{bmatrix}$$

Commented [PP17]: Answer 4.4

Problem 5

Given

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}, I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Assuming I is the identity matrix.

Find

1) IA
2) AI
3) $A(I + A)$
4) $A + IA$

Problem 5.1

$$IA$$

Since I is the identity matrix, multiplying any matrix by I yields the matrix itself.

$$IA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

Commented [PP18]: Answer 5.1

Problem 5.2

$$AI$$

Similarly, multiplying A by I yields the matrix A itself.

$$AI = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

Commented [PP19]: Answer 5.2

Problem 5.3

$$A(I + A)$$

First, find $I + A$

$$I + A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 1+6 & 0+(-2) & 0+2 \\ 0+10 & 1+(-3) & 0+1 \\ 0+(-10) & 0+5 & 1+1 \end{bmatrix}$$
$$= \begin{bmatrix} 7 & -2 & 2 \\ 10 & -2 & 1 \\ -10 & 5 & 2 \end{bmatrix}$$

Next, Compute $A(I + A)$

$$A(I+A) = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}\begin{bmatrix} 7 & -2 & 2 \\ 10 & -2 & 1 \\ -10 & 5 & 2 \end{bmatrix}$$
$$= \begin{bmatrix} (42-20-20) & (-12+4+10) & (12-2+4) \\ (70-30-10) & (-20+6+5) & (20-3+2) \\ (-70+50-10) & (20-10+5) & (-20+5+2) \end{bmatrix}$$
$$= \begin{bmatrix} 2 & 2 & 14 \\ 30 & -9 & 19 \\ -30 & 15 & -13 \end{bmatrix}$$

Commented [PP20]: Answer 5.3

Problem 5.4

$$A + IA$$

Since $IA = A$, we have $A + IA = A + A$

First, we already have $IA = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$

Then, compute $A + IA$

$$A + IA = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} + \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 6+6 & -2-2 & 2+2 \\ 10+10 & -3-3 & 1+1 \\ -10-10 & 5+5 & 1+1 \end{bmatrix}$$
$$= \begin{bmatrix} 12 & -4 & 4 \\ 20 & -6 & 2 \\ -20 & 10 & 2 \end{bmatrix}$$

Commented [PP21]: Answer 5.4